Benemérita Universidad Autónoma de Puebla
Facultad de Ciencias de la Computación

Materia: Intercomunicación y Seguridad en Redes

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Práctica 2: Uso de herramientas de consulta y propagación DNS

Objetivo

Aprender a utilizar herramientas en línea para:

- Consultar registros DNS y datos WHOIS.
- Verificar la propagación de cambios DNS a nivel mundial.
- Diagnosticar problemas relacionados con DNS.

Parte 1: Lookup Tools

1. Whois.domaintools.com

Función: Permite obtener información de registro de un dominio (propietario, registrador, fechas de creación/expiración).

Actividad detallada:

1. Abre tu navegador e ingresa la URL: https://whois.domaintools.com
2. Escribe un dominio, por ejemplo: www.buap.mx y también prueba con un segundo dominio, por ejemplo: wikipedia.org
3. Analiza la información que aparece, prestando atención a:
   - Registrador
   - Fecha de registro
   - Servidores DNS actuales
4. Guarda capturas de pantalla de ambos casos.

BUAP

DomainTools PROFILE CONNECT MONITOR SUPPORT Whois Lookup

Home > Whois Lookup > Buap.mx

Notice: Possible deprecatio

Whois Record for Buap.mx

Domain Profile

Registrar	AKKY ONLINE SOLUTIONS, S.A. DE C.V. IANA ID: – URL: http://www.akky.mx Whois Server: –
Registrar Status	
Dates	12,402 days old Created on 1992-01-31 Expires on 2026-01-30 Updated on 2025-01-27
Name Servers	NS1.BUAP.MX 148.228.1.100 (has 6 domains) NS4.BUAP.MX 148.228.1.200 (has 6 domains)
IP Address	45.60.86.125 - 7 other sites hosted on this server
IP Location	- Washington - Seattle - Incapsula Inc
ASN	AS19551 INCAPSULA, US (registered Jan 12, 2011)
Hosting History	2 changes on 2 unique name servers over 5 years

Whois Record (last updated on 2026-01-14)

```
Domain Name:        buap.mx

Created On:         1992-01-31
Expiration Date:    2026-01-30
Last Updated On:    2025-01-27
Registrar:          AKKY ONLINE SOLUTIONS, S.A. DE C.V.
URL:                http://www.akky.mx
Whois TCP URI:      whois.akky.mx
Whois Web URL:      http://www.akky.mx/herramientas/whois.jsf

Registrant:
   Name:            Benemerita Universidad Autonoma dePuebla
   City:            Puebla
   State:           Puebla
   Country:         Mexico

Administrative Contact:
   Name:            BUAP HOSTMASTER SIU-BUAP
   City:            Puebla
   State:           Puebla
   Country:         Mexico
```

Whois Record (last updated on 2026-01-14)

```
Domain Name:       buap.mx

Created On:        1992-01-31
Expiration Date:   2026-01-30
Last Updated On:   2025-01-27
Registrar:         AKKY ONLINE SOLUTIONS, S.A. DE C.V.
URL:               http://www.akky.mx
Whois TCP URI:     whois.akky.mx
Whois Web URL:     http://www.akky.mx/herramientas/whois.jsf

Registrant:
   Name:           Benemerita Universidad Autonoma dePuebla
   City:           Puebla
   State:          Puebla
   Country:        Mexico

Administrative Contact:
   Name:           BUAP HOSTMASTER SIU-BUAP
   City:           Puebla
   State:          Puebla
   Country:        Mexico

Technical Contact:
   Name:           BUAP HOSTMASTER SIU-BUAP
   City:           PUEBLA
   State:          Puebla
   Country:        Mexico

Billing Contact:
   Name:           BUAP HOSTMASTER SIU-BUAP
   City:           Puebla
   State:          Puebla
   Country:        Mexico

Name Servers:
   DNS:            ns1.buap.mx       148.228.1.100
   DNS:            ns4.buap.mx       148.228.1.200

DNSSEC DS Records:
```

Wikipedia

Whois Record for Wikipedia.org

− Domain Profile

Registrar	MarkMonitor, Inc. MarkMonitor Inc. IANA ID: 292 URL: http://www.markmonitor.com Whois Server: whois.markmonitor.com abusecomplaints@markmonitor.com (p) +1.2086851750
Registrar Status	clientDeleteProhibited, clientTransferProhibited, clientUpdateProhibited
Dates	9,132 days old Created on 2001-01-13 Expires on 2027-01-13 Updated on 2025-12-12
Name Servers	NS0.WIKIMEDIA.ORG (has 975 domains) NS1.WIKIMEDIA.ORG (has 975 domains) NS2.WIKIMEDIA.ORG (has 975 domains)
IP Address	198.35.26.96 - 37 other sites hosted on this server
IP Location	- California - San Francisco - Wikimedia Foundation Inc.
ASN	AS14907 WIKIMEDIA, US (registered Sep 27, 2006)
Domain Status	Registered And No Website
IP History	67 changes on 67 unique IP addresses over 22 years

Whois Record (last updated on 2026-01-14)

```
Domain Name: wikipedia.org
Registry Domain ID: D51687756-LROR
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2025-12-12T09:26:03+0000
Creation Date: 2001-01-13T00:12:14+0000
Registrar Registration Expiration Date: 2027-01-13T00:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibit
ed)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProh
ibited)
Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibit
ed)
Registrant Organization: Wikimedia Foundation, Inc.
Registrant Country: US
Registrant Email: Select Request Email Form at
https://domains.markmonitor.com/whois/wikipedia.org
Tech Email: Select Request Email Form at https://domains.markmonitor.com/whois/wikipe
dia.org
Name Server: ns1.wikimedia.org
Name Server: ns0.wikimedia.org
Name Server: ns2.wikimedia.org
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/

For more information on WHOIS status codes, please visit:
  https://www.icann.org/resources/pages/epp-status-codes

MarkMonitor Domain Management(TM)
Protecting companies and consumers in a digital world.

Visit MarkMonitor at https://www.markmonitor.com
Contact us at +1.8007459229
In Europe, at +44.02032062220
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```

Al consultar buap.mx, se observa que es un dominio con mucha antigüedad (creado en 1992) y su registrador es "AKKY ONLINE SOLUTIONS" (NIC México). En contraste, wikipedia.org está gestionado por "MarkMonitor Inc.", una empresa especializada en protección de marcas corporativas, y fue creado en 2001.

2. Nslookup.io

Función: Consulta registros DNS (A, MX, CNAME, TXT, etc.) sin necesidad de usar la terminal.

Actividad detallada:
1. Abre la página: https://nslookup.io
2. Ingresa un dominio, por ejemplo: : www.buap.mx y luego repite con wikipedia.org
3. Selecciona diferentes tipos de registro (A, MX, CNAME).
4. Anota:
   - IP asociada (registro A)

- Servidores de correo (MX)

5. Guarda capturas de pantalla de ambos dominios.

Buap

A records

IPv4 address	Revalidate in
45.60.113.125	1h
45.60.86.125	1h

MX records

Mail server	Priority	Revalidate in
buap-mx.mail.protection.outlook.com.	0 Primary	1h
d35d9d16f40e054b9d3ebf71042431.mail.outlook.com.	5	1h

Wikipedia

A records

IPv4 address	Revalidate in
185.15.59.224	2m 19s

AAAA records

IPv6 address	Revalidate in
2a02:ec80:300:ed1a::1	1m 33s

MX records

Mail server	Priority	Revalidate in
mx-in1001.wikimedia.org.	10 Primary	5m
mx-in2001.wikimedia.org.	10 Primary	5m

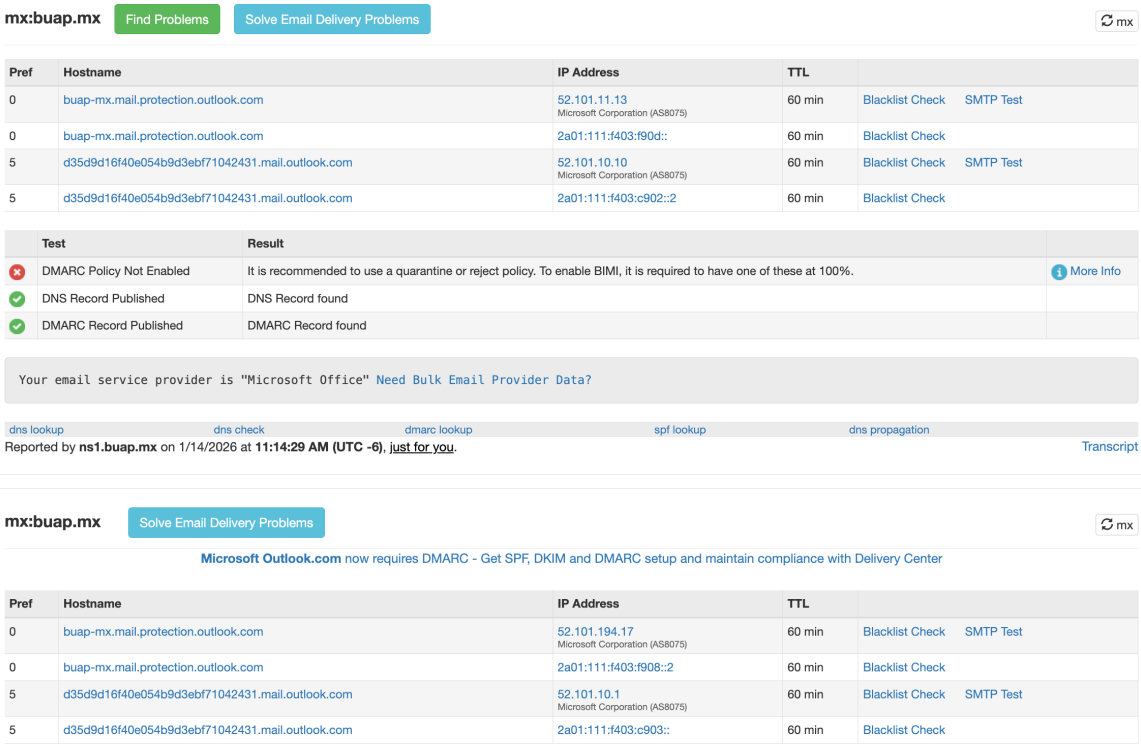
Los registros mostraron la infraestructura de cada dominio. Para la BUAP, los registros MX apuntan a buap-mx.mail.protection.outlook.com, lo que confirma que la universidad utiliza servicios de Microsoft para su correo electrónico. Para Wikipedia, los registros A y MX

apuntan a servidores internos de la fundación Wikimedia (wikimedia.org), indicando una infraestructura propia

3. Mxtoolbox.com

Función: Especializada en diagnóstico de servidores de correo y análisis de DNS.

Actividad detallada:
1. Ingresa a: https://mxtoolbox.com
2. Selecciona la herramienta “MX Lookup” e ingresa: www.buap.mx y luego wikipedia.org
3. Observa la configuración de correo y verifica si hay advertencias.
4. Captura los resultados.



	Test	Result	
✖	DMARC Policy Not Enabled	It is recommended to use a quarantine or reject policy. To enable BIMI, it is required to have one of these at 100%.	More Info
✔	DNS Record Published	DNS Record found	
✔	DMARC Record Published	DMARC Record found	

Your email service provider is "Microsoft Office" Need Bulk Email Provider Data?

dns lookup dns check dmarc lookup spf lookup dns propagation

Reported by **ns1.buap.mx** on 1/14/2026 at **11:14:07 AM (UTC -6)**, just for you. Transcript

mx:www.buap.mx

Solve Email Delivery Problems

mx

Microsoft Outlook.com now requires DMARC - Get SPF, DKIM and DMARC setup and maintain compliance with Delivery Center

Type	Domain Name	Canonical Name	TTL	
CNAME	www.buap.mx	eskew3g.x.incapdns.net	24 hrs	Blacklist Check SMTP Test

	Test	Result	
✖	DMARC Policy Not Enabled	It is recommended to use a quarantine or reject policy. To enable BIMI, it is required to have one of these at 100%.	More Info
✔	DNS Record Published	DNS Record found	
✔	DMARC Record Published	DMARC Record found	

dns lookup dns check dmarc lookup spf lookup dns propagation

Reported by **ns1.buap.mx** on 1/14/2026 at **11:13:19 AM (UTC -6)**, just for you. Transcript

mx:wikipedia.org

Find Problems Solve Email Delivery Problems

mx

PCI DSS compliance now requires DMARC - Get SPF, DKIM and DMARC setup with Delivery Center

Pref	Hostname	IP Address	TTL	
10	mx-in1001.wikimedia.org	208.80.155.102 Wikimedia Foundation Inc. (AS14907)	5 min	Blacklist Check SMTP Test
10	mx-in1001.wikimedia.org	2620:0:861:4:208:80:155:102	5 min	Blacklist Check
10	mx-in2001.wikimedia.org	208.80.153.75 Wikimedia Foundation Inc. (AS14907)	5 min	Blacklist Check SMTP Test
10	mx-in2001.wikimedia.org	2620:0:860:3:208:80:153:75	5 min	Blacklist Check

	Test	Result
✔	DNS Record Published	DNS Record found
✔	DMARC Record Published	DMARC Record found
✔	DMARC Policy Not Enabled	DMARC Quarantine/Reject policy enabled

dns lookup dns check dmarc lookup spf lookup dns propagation

Reported by **ns2.wikimedia.org** on 1/14/2026 at **11:15:36 AM (UTC -6)**, just for you. Transcript

mx:buap.mx

Solve Email Delivery Problems

mx

Pref	Hostname	IP Address	TTL	
0	buap-mx.mail.protection.outlook.com	52.101.11.13 Microsoft Corporation (AS8075)	60 min	Blacklist Check SMTP Test
0	buap-mx.mail.protection.outlook.com	2a01:111:f403:f90d::	60 min	Blacklist Check
5	d35d9d16f40e054b9d3ebf71042431.mail.outlook.com	52.101.10.10 Microsoft Corporation (AS8075)	60 min	Blacklist Check SMTP Test
5	d35d9d16f40e054b9d3ebf71042431.mail.outlook.com	2a01:111:f403:c902::2	60 min	Blacklist Check

	Test	Result

Esta herramienta reveló diferencias en la configuración de seguridad. Mientras que wikipedia.org pasó todas las pruebas con éxito (incluyendo política DMARC en modo "Quarantine/Reject") , el dominio buap.mx mostró una advertencia indicando "DMARC Policy Not Enabled", lo cual es una oportunidad de mejora en la seguridad de su correo.

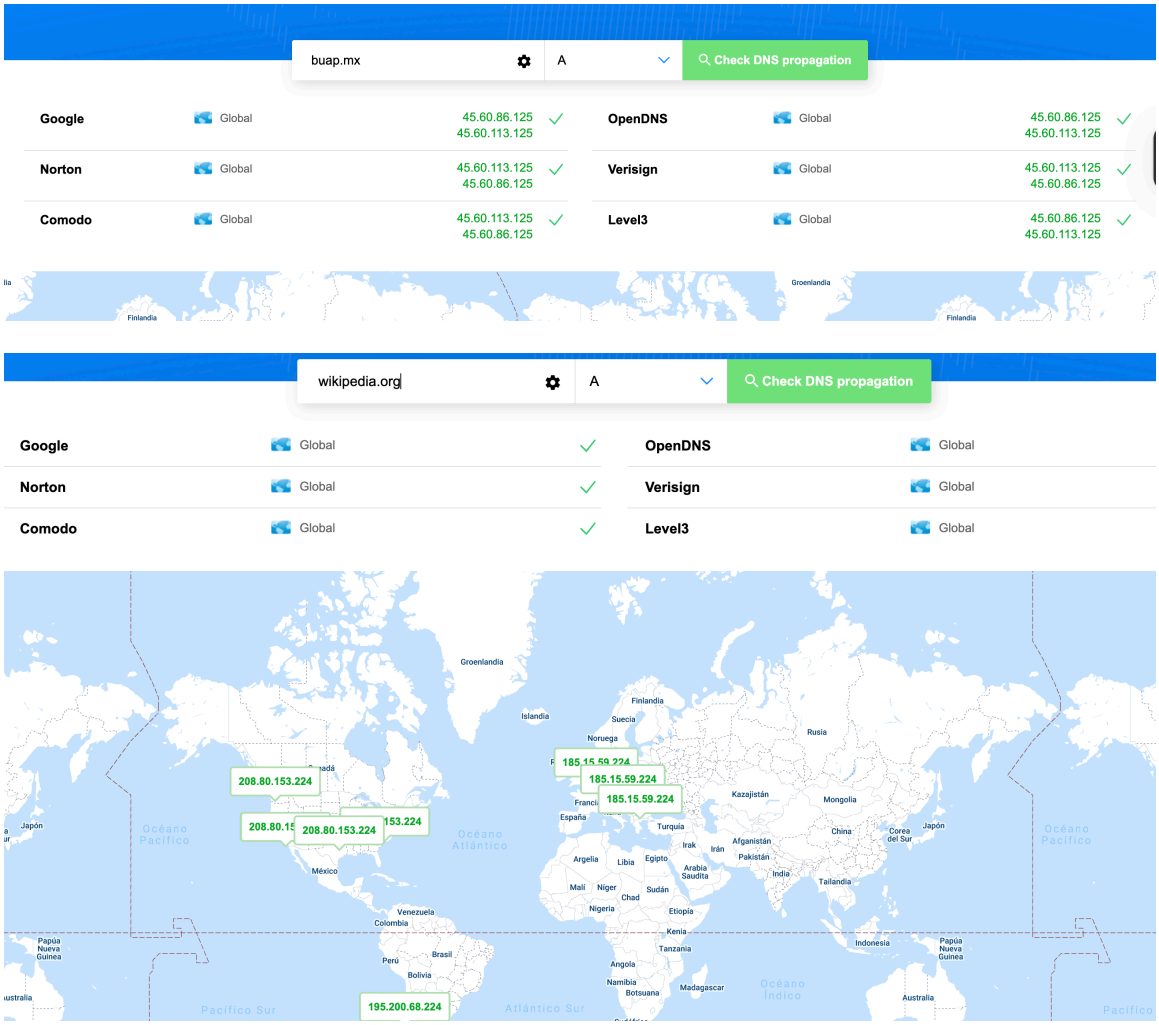
Parte 2: Propagación de DNS

4. Dnsmap.io

Función: Muestra cómo se resuelve un dominio desde diferentes ubicaciones en el mundo.

Actividad:

1. Ingresa a: https://dnsmap.io
2. Consulta: www.buap.mx y wikipedia.org
3. Observa las IP devueltas en cada país.
4. Compara si todas son iguales o hay diferencias (posible propagación incompleta).



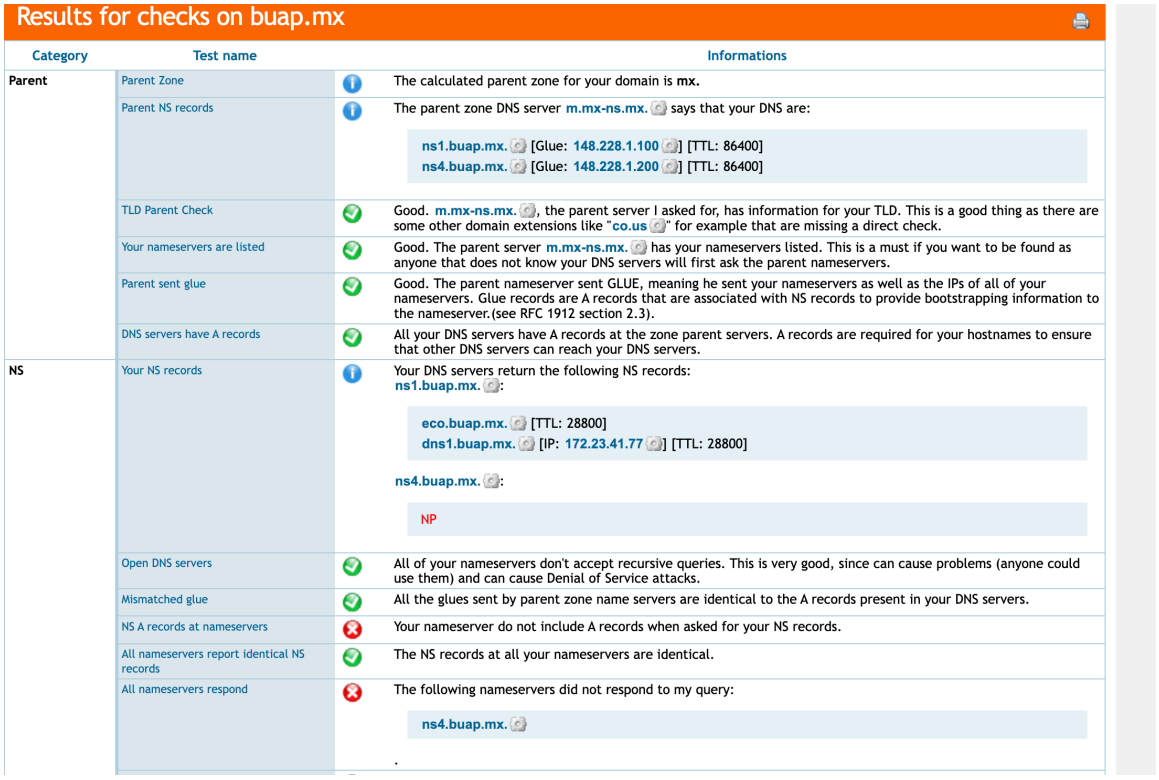
 BUAP: Muestra una **propagación estática**. Las direcciones IP (como 45.60.113.125) son consistentes en casi todo el mundo (EE.UU., Europa, Asia), lo que sugiere que el tráfico se dirige a un punto centralizado o proxy inverso.

 Wikipedia: Muestra una **propagación basada en ubicación (Anycast/CDN)**. Dependiendo del país desde donde se consulte, la IP cambia (ej. 185.15.59.224 en Europa vs 208.80.154.224 en EE.UU.). Esto confirma que Wikipedia utiliza una Red de Entrega de Contenidos para servir la información desde el servidor más cercano al usuario.

5. Dnsqueries.com Domain Check

Función: Revisión general de configuración DNS, incluyendo MX, NS, y registros A.

Actividad:
1. Ingresa a: https://dnsqueries.com/en/domain_check.php
2. Consulta : www.buap.mx y wikipedia.org
3. Anota si hay advertencias o errores.



Results for checks on buap.mx

Category	Test name	Informations
Parent	Parent Zone	The calculated parent zone for your domain is **mx.**
	Parent NS records	The parent zone DNS server **m.mx-ns.mx.** says that your DNS are: **ns1.buap.mx.** [Glue: **148.228.1.100**] [TTL: 86400] **ns4.buap.mx.** [Glue: **148.228.1.200**] [TTL: 86400]
	TLD Parent Check	Good. **m.mx-ns.mx.**, the parent server I asked for, has information for your TLD. This is a good thing as there are some other domain extensions like "**co.us**" for example that are missing a direct check.
	Your nameservers are listed	Good. The parent server **m.mx-ns.mx.** has your nameservers listed. This is a must if you want to be found as anyone that does not know your DNS servers will first ask the parent nameservers.
	Parent sent glue	Good. The parent nameserver sent GLUE, meaning he sent your nameservers as well as the IPs of all of your nameservers. Glue records are A records that are associated with NS records to provide bootstrapping information to the nameserver.(see RFC 1912 section 2.3).
	DNS servers have A records	All your DNS servers have A records at the zone parent servers. A records are required for your hostnames to ensure that other DNS servers can reach your DNS servers.
NS	Your NS records	Your DNS servers return the following NS records: **ns1.buap.mx.**: **eco.buap.mx.** [TTL: 28800] **dns1.buap.mx.** [IP: **172.23.41.77**] [TTL: 28800] **ns4.buap.mx.**: NP
	Open DNS servers	All of your nameservers don't accept recursive queries. This is very good, since can cause problems (anyone could use them) and can cause Denial of Service attacks.
	Mismatched glue	All the glues sent by parent zone name servers are identical to the A records present in your DNS servers.
	NS A records at nameservers	Your nameserver do not include A records when asked for your NS records.
	All nameservers report identical NS records	The NS records at all your nameservers are identical.
	All nameservers respond	The following nameservers did not respond to my query: **ns4.buap.mx.**

Nameserver name validity	✓	All of the NS records that your nameservers report seem valid hostnames.
Number of nameservers	✓	You have 2 nameservers. You must have at least 2 nameservers and no more than 7.
Lame nameservers	✓	All the nameservers listed at the parent servers answer authoritatively for your domain.
Missing (stealth) nameservers	✗	I have detected 2 stealth nameservers: **eco.buap.mx.** **dns1.buap.mx.** . These are listed in your nameservers but are missing in the parent zone nameservers. Those nameservers are not included in these tests, so you have to check them manually.
Missing (stealth) nameservers 2	✗	I have detected 2 stealth nameservers: **ns1.buap.mx.** **ns4.buap.mx.** . These are listed in the parent zone nameservers but are missing in your nameservers
No CNAMEs for domain	✓	There are no CNAMEs for **buap.mx.**. RFC1912 2.4 and RFC2181 10.3 state that there should be no CNAMEs if an NS (or any other) record is present.
Nameservers on separate class C's	✗	Your nameservers are on the same Class C IP range. This is very bad if you want to be found in the case of outage, or even worst, problems!
All NS IPs public	✓	All of your NS records appear to use public IPs.
TCP Allowed	✗	These servers are not accepting TCP connections: **172.23.41.77**: Connection timed out [110] TCP connections are occasionally used instead of UDP connections and can be blocked by firewalls. This can cause hard-to-diagnose problems.
Your SOA records	ℹ	Your DNS servers return the following SOA records: **ns1.buap.mx.**: **dns1.buap.mx.** please_set_email.absolutely.nowhere. 4832 10800 3600 2419200 900. [TTL: 28800] **ns4.buap.mx.**: NP
All same SOA	✓	All your nameserver respond with the same SOA record, wich is a great thing! The SOA record is: Primary nameserver: **dns1.buap.mx.** Hostmaster E-mail address: please_set_email.absolutely.nowhere. Serial Number: 4832 Refresh: 10800 Retry: 3600 Expire: 2419200 Default TTL: 900.

		same identifier for the datas.
SOA Primary NS	✗	Not all your nameservers agree on the identification of the primary nameserver or it isn't listed in the parent zone nameserver.
Hostmaster Email	✓	All your nameservers state that your hostmaster E-Mail address is **please_set_email.absolutely.nowhere.**.
Serial Format	✗	Your SOA serial number is **4832**. It doesn't appear to be in therecommended format of YYYYMMDDnn, you should change it accordingly.
REFRESH	✗	The SOA REFRESH value determines how often secondary nameservers check with the master nameserver for updates.Your SOA REFRESH value is **10800** seconds which is very high (about 3600-7200 seconds is good althought RFC1912 2.2 recommends a value between 1200 to 43200 seconds).
RETRY	✓	The retry value is the amount of time your secondary nameservers will wait to contact the master nameserver again if the last attempt failed.Your SOA RETRY interval is **3600** seconds and it seems normal (120-7200 seconds is ok).
EXPIRE	✓	The expire value is how long a secondary nameserver will wait before considering its DNS data stale if it can't reach the primary nameserver. Your SOA EXPIRE value is **2419200** seconds which seems normal (as suggested by RFC1912 a value between 1209600 to 2419200 seconds is good).
Default TTL	!	The default TTL is used for negative caching, meaning that all the queries that don't have a valid response are cached for this amount of seconds. Your SOA DEFAULT TTL is: **900** seconds which is very low (as suggested by RFC2308 a value between 3600 and 86400 seconds is ok). This means that a failed query for a not-existant domain will be cached for a little time, and, after this amount of seconds, will be re-done.
Your MX records	i	Your DNS servers return the following MX records: **ns1.buap.mx.**: 0 **buap-mx.mail.protection.outlook.com.** [TTL: 3600] 5 **d35d9d16f40e054b9d3ebf71042431.mail.outlook.com.** [TTL: 3600] **ns4.buap.mx.**: NP
Multiple MX records	✓	You have multiple MX records and this is a very good thing! When one MX server is down the others can continue to accept mail.
Invalid characters	✓	It seems that all of your MX records use valid hostnames, without any invalid characters.
All MX IPs public	!	Your NS don't return their IPs when looking explicitly for MX records.
MX records are not CNAMEs	✓	None of the lookups of your MX records did return CNAMEs.
MX A lookups have no CNAMEs	✓	Looking up for the A records of your MX servers i did not detect problems.
MX is host name, not IP	✓	All the MX records retrieved are host names. Using IP addresses in MX records is notallowed.
Differing MX-A records	✓	I have searched for differing IPs for your MX records between what are declaring your NS and the authoritative NS for the MX records. All was fine.
Duplicate MX records	✓	I have checked and all your MX records are pointing to different IPs.
Reverse DNS entries for MX records	✓	All the IPs of your MX records have PTR entries, meaning it is possible to know their hostnames from the IP address. Many mailservers, accordingly to RFC1912 2.1, will not accept mail from mailservers with no PTR (reverse DNS) entry. The reverse DNS entries are **buap-mx.mail.protection.outlook.com.** -> **52.101.11.9** -> **mail-sn1pr02cu00101.inbound.protection.outlook.com**

Mail server host name in greeting	✗	The configuration of your mail servers and your DNS are not ok! The report of the test is: **buap-mx.mail.protection.outlook.com.** -> **BN2PEPF00004FBB.mail.protection.outlook.com** -> N.C. **d35d9d16f40e054b9d3ebf71042431.mail.outlook.com.** -> **CH2PEPF0000009C.mail.protection.outlook.com** -> N.C. Spam recognition software and RFC821 4.3 (also RFC2821 4.3.1) state that the hostname given in the SMTP greeting MUST have an A record pointing back to the same server.
Acceptance of NULL <> sender	✓	All of your mailservers accept mail from "<>". RFC1123 5.2.9 requires all mailservers to receive mail from this kind of address, which is used in reject/bounce messages and return receipts. The report of the test is: **buap-mx.mail.protection.outlook.com.**: Accepts null sender **d35d9d16f40e054b9d3ebf71042431.mail.outlook.com.**: Accepts null sender
Acceptance of postmaster address	✗	Not all of your mailservers accept mail to postmaster@buap.mx.. RFC822 6.3, RFC1123 5.2.7, and RFC2821 4.5.1 require all mailservers to accept mail to this kind of address. The report of the test is: **buap-mx.mail.protection.outlook.com.**: Does not accept mail to postmaster@buap.mx. **d35d9d16f40e054b9d3ebf71042431.mail.outlook.com.**: Does not accept mail to postmaster@buap.mx.
Acceptance of abuse address	✗	Not all of your mailservers accept mail to postmaster@buap.mx.. RFC822 6.3, RFC1123 5.2.7, and RFC2821 4.5.1 require all mailservers to accept mail to this kind of address. The report of the test is: **buap-mx.mail.protection.outlook.com.**: Does not accept mail to abuse@buap.mx. **d35d9d16f40e054b9d3ebf71042431.mail.outlook.com.**: Does not accept mail to abuse@buap.mx.
Acceptance of domain literals	✗	Not all of your mailservers accept mail to postmaster@[ip_address] (Literal format). RFC1123 5.2.17 require all mailservers to accept mail to this kind of address. This is a common problem and actually can be ignored.The report of the test is: **buap-mx.mail.protection.outlook.com.**: Does not accept mail to postmaster@[ip_address] **d35d9d16f40e054b9d3ebf71042431.mail.outlook.com.**: Does not accept mail to postmaster@[ip_address]
Open relay test	✓	All of your mailservers are closed to external domain relaying.The report of the test is: **buap-mx.mail.protection.outlook.com.**: Is not an open relay **d35d9d16f40e054b9d3ebf71042431.mail.outlook.com.**: Is not an open relay
SPF record	!	You don't have a SPF record for the domain **buap.mx.**, meaning that you are not using the protection given from this kind of technology. This is only a warning, but please consider in implementing it!
WWW Record	!	I have asked your DNS server for **www.buap.mx.** but i did not receive an IP address (maybe i received a CNAME...), however these are the records i received: **www.buap.mx.** = CNAME **eskew3g.x.incapdns.net.**
All WWW IPs public	–	I have no ip addresses to check
CNAME Lookup	!	There is one or more CNAMEs record pointing to www.buap.mx.. This can cause extra bandwidth usage since the resolution of **www.buap.mx.** is done in multiple steps. However this is only a warning!

wikipedia

Results for checks on wikipedia.org

Category	Test name		Informations
Parent	Parent Zone	ℹ	The calculated parent zone for your domain is **org.**
	Parent NS records	ℹ	The parent zone DNS server **d0.org.afilias-nst.org.** says that your DNS are: **ns2.wikimedia.org.** [Glue: **198.35.27.27**] [TTL: 3600] **ns1.wikimedia.org.** [Glue: **208.80.153.231**] [TTL: 3600] **ns0.wikimedia.org.** [Glue: **208.80.154.238**] [TTL: 3600]
	TLD Parent Check	✓	Good. **d0.org.afilias-nst.org.**, the parent server I asked for, has information for your TLD. This is a good thing as there are some other domain extensions like "**co.us**" for example that are missing a direct check.
	Your nameservers are listed	✓	Good. The parent server **d0.org.afilias-nst.org.** has your nameservers listed. This is a must if you want to be found as anyone that does not know your DNS servers will first ask the parent nameservers.
	Parent sent glue	✓	Good. The parent nameserver sent GLUE, meaning he sent your nameservers as well as the IPs of all of your nameservers. Glue records are A records that are associated with NS records to provide bootstrapping information to the nameserver.(see RFC 1912 section 2.3).
	DNS servers have A records	✓	All your DNS servers have A records at the zone parent servers. A records are required for your hostnames to ensure that other DNS servers can reach your DNS servers.
NS	Your NS records	ℹ	Your DNS servers return the following NS records: **ns2.wikimedia.org.**: **ns0.wikimedia.org.** [TTL: 86400] **ns1.wikimedia.org.** [TTL: 86400] **ns2.wikimedia.org.** [TTL: 86400] **ns1.wikimedia.org.**: **ns0.wikimedia.org.** [TTL: 86400] **ns1.wikimedia.org.** [TTL: 86400] **ns2.wikimedia.org.** [TTL: 86400] **ns0.wikimedia.org.**: **ns0.wikimedia.org.** [TTL: 86400] **ns1.wikimedia.org.** [TTL: 86400] **ns2.wikimedia.org.** [TTL: 86400]
	Open DNS servers	✓	All of your nameservers don't accept recursive queries. This is very good, since can cause problems (anyone could use them) and can cause Denial of Service attacks.
	Mismatched glue	✓	All the glues sent by parent zone name servers are identical to the A records present in your DNS servers.
	NS A records at nameservers	✗	Your nameserver do not include A records when asked for your NS records.
	All nameservers report identical NS records	✓	The NS records at all your nameservers are identical.
	All nameservers respond	✓	All of your nameservers listed at the parent nameservers responded.
	Nameserver name validity	✓	All of the NS records that your nameservers report seem valid hostnames.
	Number of nameservers	✓	You have 3 nameservers. You must have at least 2 nameservers and no more than 7.

Lame nameservers	✓	All the nameservers listed at the parent servers answer authoritatively for your domain.
Missing (stealth) nameservers	✓	All of your nameservers are listed at the parent zone servers.
Missing (stealth) nameservers 2	✓	All of the nameservers listed at the parent zone nameservers are listed as NS records at your nameservers.
No CNAMEs for domain	✓	There are no CNAMEs for **wikipedia.org.**. RFC1912 2.4 and RFC2181 10.3 state that there should be no CNAMEs if an NS (or any other) record is present.
Nameservers on separate class C's	✗	Your nameservers are on the same Class C IP range. This is very bad if you want to be found in the case of outage, or even worst, problems!
All NS IPs public	✓	All of your NS records appear to use public IPs.
TCP Allowed	✓	All your DNS servers allow TCP connections. TCP connections are occasionally used instead of UDP connections and can be blocked by firewalls. This can cause hard-to-diagnose problems.
Your SOA records	ℹ	Your DNS servers return the following SOA records: **ns2.wikimedia.org.**: **ns0.wikimedia.org.** **hostmaster.wikimedia.org.** 2025120109 43200 7200 1209600 3600. [TTL: 3600] **ns1.wikimedia.org.**: **ns0.wikimedia.org.** **hostmaster.wikimedia.org.** 2025120109 43200 7200 1209600 3600. [TTL: 3600] **ns0.wikimedia.org.**: **ns0.wikimedia.org.** **hostmaster.wikimedia.org.** 2025120109 43200 7200 1209600 3600. [TTL: 3600]
All same SOA	✓	All your nameserver respond with the same SOA record, wich is a great thing! The SOA record is: Primary nameserver: **ns0.wikimedia.org.** Hostmaster E-mail address: **hostmaster.wikimedia.org.** Serial Number: 2025120109 Refresh: 43200 Retry: 7200 Expire: 1209600 Default TTL: 3600.
Same Serial Number	✓	All your nameservers agree that your SOA serial number is **2025120109**. That means that all your nameservers are using the same identifier for the datas.
SOA Primary NS	✓	All your nameservers indicate **ns0.wikimedia.org.** as your primary nameserver, which is listed in the parent zone nameserver.
Hostmaster Email	✓	All your nameservers state that your hostmaster E-Mail address is **hostmaster.wikimedia.org.**.
Serial Format	✓	Your SOA serial number is **2025120109**. It appears to be in the format of YYYYMMDDnn (recommended), where 'nn' is the revision. Your DNS was last updated on 1 December 2025 and was revision 09.
REFRESH	✗	The SOA REFRESH value determines how often secondary nameservers check with the master nameserver for updates.Your SOA REFRESH value is **43200** seconds which is very high (about 3600-7200 seconds is good althought RFC1912 2.2 recommends a value between 1200 to 43200 seconds).
RETRY	✓	The retry value is the amount of time your secondary nameservers will wait to contact the master nameserver again if the last attempt failed.Your SOA RETRY interval is **7200** seconds and it seems normal (120-7200 seconds is ok).
EXPIRE	✓	The expire value is how long a secondary nameserver will wait before considering its DNS data stale if it can't reach the primary nameserver. Your SOA EXPIRE value is **1209600** seconds which seems normal (as suggested by RFC1912 a value between 1209600 to 2419200 seconds is good).

MX	Your MX records		Your DNS servers return the following MX records: **ns2.wikimedia.org.**: 10 **mx-in1001.wikimedia.org.** [TTL: 300] 10 **mx-in2001.wikimedia.org.** [TTL: 300] **ns1.wikimedia.org.**: 10 **mx-in1001.wikimedia.org.** [TTL: 300] 10 **mx-in2001.wikimedia.org.** [TTL: 300] **ns0.wikimedia.org.**: 10 **mx-in1001.wikimedia.org.** [TTL: 300] 10 **mx-in2001.wikimedia.org.** [TTL: 300]
	Multiple MX records		You have multiple MX records and this is a very good thing! When one MX server is down the others can continue to accept mail.
	Invalid characters		It seems that all of your MX records use valid hostnames, without any invalid characters.
	All MX IPs public		Your NS don't return their IPs when looking explicitly for MX records.
	MX records are not CNAMEs		None of the lookups of your MX records did return CNAMEs.
	MX A lookups have no CNAMEs		Looking up for the A records of your MX servers i did not detect problems.
	MX is host name, not IP		All the MX records retrieved are host names. Using IP addresses in MX records is notallowed.
	Differing MX-A records		I have searched for differing IPs for your MX records between what are declaring your NS and the authoritative NS for the MX records. All was fine.
	Duplicate MX records		I have checked and all your MX records are pointing to different IPs.
	Reverse DNS entries for MX records		All the IPs of your MX records have PTR entries, meaning it is possible to know their hostnames from the IP address. Many mailservers, accordingly to RFC1912 2.1, will not accept mail from mailservers with no PTR (reverse DNS) entry. The reverse DNS entries are **mx-in1001.wikimedia.org.** -> **208.80.155.102** -> **mx-in1001.wikimedia.org** **mx-in2001.wikimedia.org.** -> **208.80.153.75** -> **mx-in2001.wikimedia.org**
MAIL	Connect to mail servers		I have connected successfully to all of your mailservers. **mx-in1001.wikimedia.org.**: Connected with greeting: **mx-in1001.wikimedia.org** **mx-in2001.wikimedia.org.**: Connected with greeting: **mx-in2001.wikimedia.org**
	Mail server host name in greeting		Your mail servers respond with the hostname in the greeting message and everything works fine. The report of the test is: **mx-in1001.wikimedia.org.** -> **mx-in1001.wikimedia.org** -> 208.80.155.102 -> **mx-in1001.wikimedia.org** **mx-in2001.wikimedia.org.** -> **mx-in2001.wikimedia.org** -> 208.80.153.75 -> **mx-in2001.wikimedia.org** This is very good, since spam recognition software and RFC821 4.3 (also RFC2821 4.3.1) state that the hostname given in the SMTP greeting MUST have an A record pointing back to the same server.
Acceptance of NULL <> sender			All of your mailservers accept mail from "<>". RFC1123 5.2.9 requires all mailservers to receive mail from this kind of address, which is used in reject/bounce messages and return receipts. The report of the test is: **mx-in1001.wikimedia.org.**: Accepts null sender **mx-in2001.wikimedia.org.**: Accepts null sender
Acceptance of postmaster address			All of your mailservers accept mail to postmaster@wikipedia.org.. RFC822 6.3, RFC1123 5.2.7, and RFC2821 4.5.1 require all mailservers to accept mail to this kind of address. The report of the test is: **mx-in1001.wikimedia.org.**: Accepts mail to postmaster@wikipedia.org. **mx-in2001.wikimedia.org.**: Accepts mail to postmaster@wikipedia.org.
Acceptance of abuse address			All of your mailservers accept mail to postmaster@wikipedia.org.. RFC822 6.3, RFC1123 5.2.7, and RFC2821 4.5.1 require all mailservers to accept mail to this kind of address. The report of the test is: **mx-in1001.wikimedia.org.**: Accepts mail to abuse@wikipedia.org. **mx-in2001.wikimedia.org.**: Accepts mail to abuse@wikipedia.org.
Acceptance of domain literals			All of your mailservers accept mail to postmaster@[ip_address] (Literal format). RFC1123 5.2.17 require all mailservers to accept mail to this kind of address. The report of the test is: **mx-in1001.wikimedia.org.**: Accepts mail to postmaster@[ip_address] **mx-in2001.wikimedia.org.**: Accepts mail to postmaster@[ip_address]
Open relay test			All of your mailservers are closed to external domain relaying.The report of the test is: **mx-in1001.wikimedia.org.**: Is not an open relay **mx-in2001.wikimedia.org.**: Is not an open relay
SPF record			You have an SPF record And this is very very good, as it will help to stop spammers using the domain **wikipedia.org.** in their activities.Your SPF record is: "v=spf1 include:_cidrs.wikimedia.org -all". Please note that i am not checking if it is a valid SPF record...
WWW Record			I have asked your DNS server for **www.wikipedia.org.** but i did not receive an IP address (maybe i received a CNAME...), however these are the records i received: **www.wikipedia.org.** = CNAME **dyna.wikimedia.org.**
All WWW IPs public			I have no ip addresses to check
CNAME Lookup			There is one or more CNAMEs record pointing to www.wikipedia.org.. This can cause extra bandwidth usage since the resolution of **www.wikipedia.org.** is done in multiple steps. However this is only a warning!

Realizó un análisis profundo de la zona DNS. Para la BUAP, validó que los servidores de nombres (NS) están configurados correctamente para no aceptar consultas recursivas (previniendo ataques DDoS), aunque lanzó una advertencia menor sobre el valor de actualización (Refresh) del registro SOA

6. Viewdns.info Propagation

Función: Comprobación visual de la propagación de un dominio en múltiples servidores DNS.

Actividad:
1. Ingresa a: https://viewdns.info/propagation
2. Consulta : www.buap.mx y wikipedia.org
3. Verifica si todos los servidores muestran la misma IP.

SERVER LOCATION	RECORDS RETURNED	STATUS
Atlanta, USA	45.60.113.125,45.60.86.125	✓
Beauharnois, Canada	45.60.113.125,45.60.86.125	✓
Hong Kong, China	45.60.113.125,45.60.86.125	✓
Johannesburg, South Africa	45.60.113.125,45.60.86.125	✓
London, United Kingdom	45.60.113.125,45.60.86.125	✓
Los Angeles, USA	45.60.113.125,45.60.86.125	✓
Mumbai, India	45.60.113.125,45.60.86.125	✓
New York, USA	45.60.113.125,45.60.86.125	✓
Oslo, Norway	45.60.113.125,45.60.86.125	✓
Paris, France	45.60.113.125,45.60.86.125	✓
Rotterdam, Netherlands	45.60.113.125,45.60.86.125	✓
Santiago, Chile	45.60.113.125,45.60.86.125	✓
Seattle, USA	45.60.113.125,45.60.86.125	✓

wikipedia

SERVER LOCATION	RECORDS RETURNED	STATUS
Atlanta, USA	208.80.154.224	✗
Beauharnois, Canada	208.80.153.224	✓
Hong Kong, China	103.102.166.224	✗
Johannesburg, South Africa	185.15.58.224	✗
London, United Kingdom	185.15.59.224	✗
Los Angeles, USA	208.80.153.224	✓
Mumbai, India	103.102.166.224	✗
New York, USA	208.80.154.224	✗
Oslo, Norway	185.15.59.224	✗
Paris, France	185.15.58.224	✗
Rotterdam, Netherlands	185.15.59.224	✗
Santiago, Chile	195.200.68.224	✗
Seattle, USA	208.80.153.224	✓

7. Whatsmydns.net

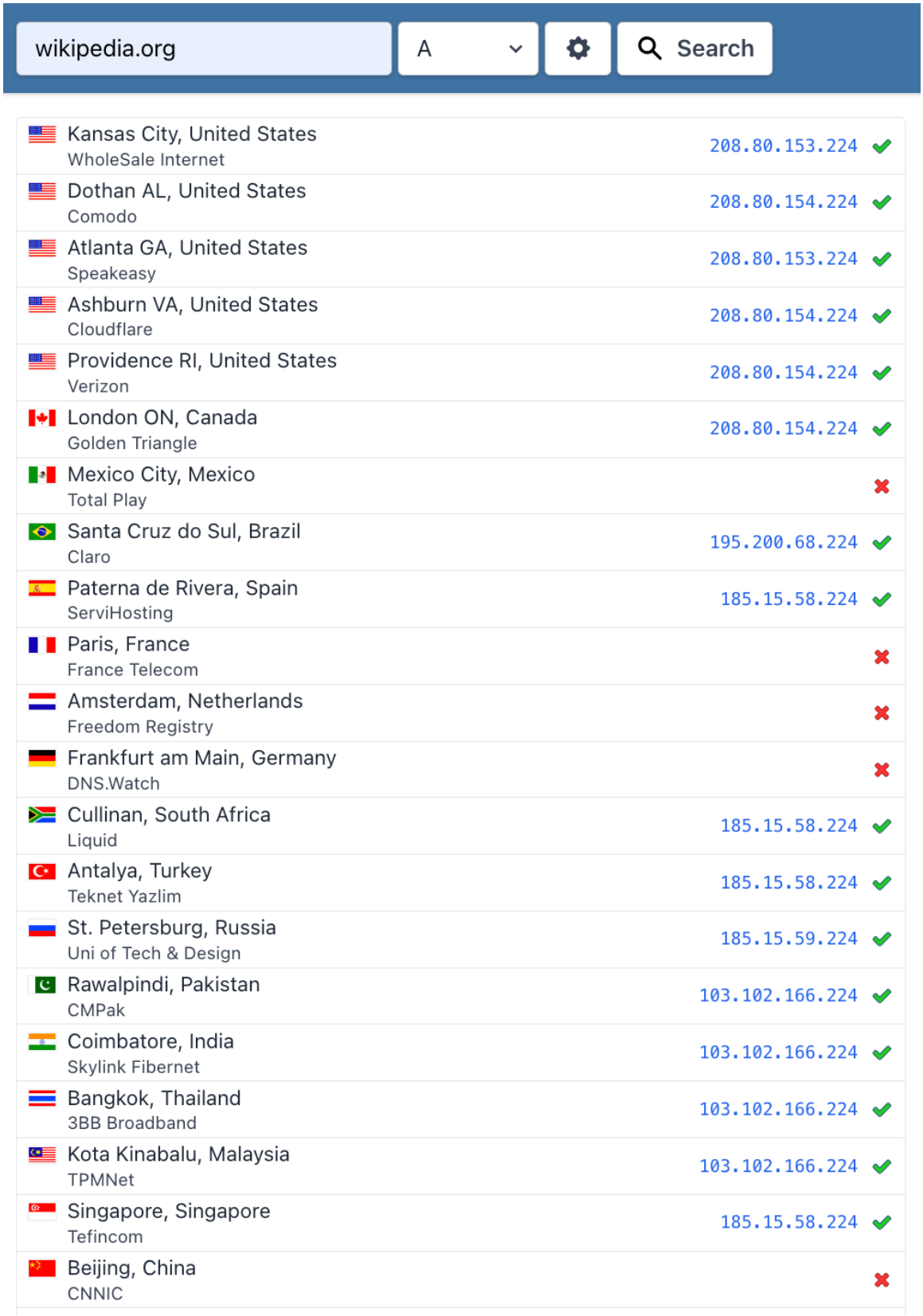
Función: Herramienta muy visual para ver propagación por tipo de registro.

Actividad:
1. Ingresa a: https://www.whatsmydns.net
2. Escribe : www.buap.mx y selecciona el tipo de registro A.
3. Repite con wikipedia.org.
4. Observa si todos los puntos del mapa muestran la misma IP

buap.mx | A | Search

Kansas City, United States WholeSale Internet	45.60.113.125 45.60.86.125	✔
Dothan AL, United States Comodo	45.60.113.125 45.60.86.125	✔
Atlanta GA, United States Speakeasy	45.60.113.125 45.60.86.125	✔
Ashburn VA, United States Cloudflare	45.60.113.125 45.60.86.125	✔
Providence RI, United States Verizon		✖
London ON, Canada Golden Triangle	45.60.113.125 45.60.86.125	✔
Mexico City, Mexico Total Play		✖
Santa Cruz do Sul, Brazil Claro	45.60.113.125 45.60.86.125	✔
Paterna de Rivera, Spain ServiHosting	45.60.113.125 45.60.86.125	✔
Paris, France France Telecom	45.60.113.125 45.60.86.125	✔
Amsterdam, Netherlands Freedom Registry	45.60.113.125 45.60.86.125	✔
Frankfurt am Main, Germany DNS.Watch		✖
Cullinan, South Africa Liquid	45.60.113.125 45.60.86.125	✔
Antalya, Turkey Teknet Yazlim	45.60.113.125 45.60.86.125	✔
St. Petersburg, Russia Uni of Tech & Design	45.60.113.125 45.60.86.125	✔
Rawalpindi, Pakistan CMPak	45.60.113.125 45.60.86.125	✔
Coimbatore, India Skylink Fibernet	45.60.113.125 45.60.86.125	✔

wikipedia


wikipedia.org
A
Search
Kansas City, United States
WholeSale Internet
208.80.153.224
Dothan AL, United States
Comodo
208.80.154.224
Atlanta GA, United States
Speakeasy
208.80.153.224
Ashburn VA, United States
Cloudflare
208.80.154.224
Providence RI, United States
Verizon
208.80.154.224
London ON, Canada
Golden Triangle
208.80.154.224
Mexico City, Mexico
Total Play
Santa Cruz do Sul, Brazil
Claro
195.200.68.224
Paterna de Rivera, Spain
ServiHosting
185.15.58.224
Paris, France
France Telecom
Amsterdam, Netherlands
Freedom Registry
Frankfurt am Main, Germany
DNS.Watch
Cullinan, South Africa
Liquid
185.15.58.224
Antalya, Turkey
Teknet Yazlim
185.15.58.224
St. Petersburg, Russia
Uni of Tech & Design
185.15.59.224
Rawalpindi, Pakistan
CMPak
103.102.166.224
Coimbatore, India
Skylink Fibernet
103.102.166.224
Bangkok, Thailand
3BB Broadband
103.102.166.224
Kota Kinabalu, Malaysia
TPMNet
103.102.166.224
Singapore, Singapore
Tefincom
185.15.58.224
Beijing, China
CNNIC

8. Nexcess DNS Checker

Función: Permite revisar propagación y hacer consultas rápidas de registros DNS.

Actividad:
1. Ingresa a: https://www.nexcess.net/tools/dns-checker/
2. Realiza una búsqueda de un registro A o MX para: www.buap.mx y wikipedia.org
3. Documenta los resultados.

Conclusion: "La herramienta que me pareció más útil fue **Whatsmydns.net**. Su interfaz basada en un mapamundi permite comprender instantáneamente el concepto de propagación DNS. Gracias a ella, pude visualizar claramente la diferencia entre una infraestructura centralizada (como la de la BUAP, que mostraba la misma IP globalmente) y una distribuida (como Wikipedia, que cambiaba de IP según la región), algo que hubiera sido difícil de notar solo mirando listas de texto."

Entrega de la práctica

• Documento en Word o PDF con capturas de pantalla de cada herramienta (para los dos dominios probados).
• Breve explicación de lo que encontraste en cada consulta.
• Conclusión sobre cuál herramienta te pareció más útil y por qué.